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Summary

Full-Stack AI Engineer who enjoys the craft of shipping end-to-end AI products, from document understanding and RAG pipelines to agentic AI systems, desktop apps to web platforms, and air-gapped deployments to cloud-based enterprise solutions. At NeoALI, shipping on-device AI desktop applications used by 100+ users and building web-based AI solutions for office workflows, currently piloting with enterprise clients.

Feel free to chat with my AI assistant at kc-bibek.com.np.

Core Expertise

• AI Product Development • Agentic AI Systems • LLM Serving • Document Intelligence & RAG • Backend API Architecture • Enterprise & Air-Gapped Deployment • Desktop & Web Apps • Technical Leadership

Experience

AI Researcher & On-device Team Lead

NeoALI Co. Ltd, Korea

Aug 2023 – Present

AI Products

- Technical lead of a 3-member on-device AI team, driving architecture decisions, deployment strategy, and cross-product engineering initiatives.
- Built and shipped [SamuGen](#) and [SamuGen Plug-in](#), on-device AI desktop writing assistants used by 100+ users across air-gapped and B2C environments, supporting document-aware AI writing, Doc QnA, and HWP/HWPX/DOCX import/export, with a production-ready web version for on-premise deployment.
- Architected the backend of NeoClaw, an on-premise agentic proposal evaluation platform orchestrating scoring, summarization, synthesis, and ranking subagents, with QnA for querying AI scores.
- Designed and developed SlideGen, an on-premise LLM-powered slide generation and editing platform that creates presentations from uploaded documents with support for PPTX export.

Agentic Systems

- Built a unified agentic chat interface as the entrypoint across all on-premise AI products, with workflow orchestration that routes user intent to appropriate backends including proposal evaluation, RAG QnA, document translation, and other services.
- Engineered an internal service that sandboxes the pi harness inside per-user/per-session OpenShell containers with default-deny egress, giving an enterprise RAG chatbot secure, isolated agentic task execution via a streaming API.

AI Infrastructure

- Designed backend architecture using FastAPI, Express, Redis, SSE, authentication services, and API gateways, serving as shared infrastructure across multiple AI products.
- Operated shared LLM-serving infrastructure using vLLM, exposing OpenAI-compatible APIs for language models, embeddings, and reranking, serving as shared backend across 5 internal products. For on-device deployment, switched between IPEX-LLM Ollama, Llama-cpp, and OpenVINO.
- Contributed core components to ALIParse (in-house library), including layout analysis, reading-order ranking, document parsing, semantic retrieval, and RAG pipelines.

Enterprise Deployment

- Led deployment of AI desktop applications to 50+ air-gapped military notebooks, implementing offline license activation, model distribution, code obfuscation, and enterprise installation workflows.
- Built CI/CD pipelines, cross-platform packaging (Windows/macOS), monitoring dashboards, Slack-based alerting, load and stress testing, product download portals, and SMTP email workflows.

Research Intern

Robust Intelligence & Robotics Lab, KAIST

Jul 2022 - Dec 2022 (6 months)

- For a single moving camera setup, developed a combined framework for real-time 6DoF Pose Estimation and Scene Graph Generation (SGG) in Robotic Manipulation Task using [Cosypose](#) framework for 6DoF pose and [Neural-Motif](#) framework for SGG and performed experiments on custom cup datasets. [\[Demo\]](#)
- For 2 fixed cameras setup, developed a probabilistic multi-view object-pose estimation framework that (i) associates multiple estimation results with scene graphs, (ii) combines pose distributions from singleview based estimators, and (iii) constructs a unified scene graph by predicting a unified pose distribution per object using MC dropout. More details in [draft-paper](#).

Backend Developer Intern

Bisonai, Seoul

Jan 2022 - Feb 2022 (2 months)

- Built data pipelines (DAGs) in Airflow to collect DeFi data from various sources like Uniswap, Coingecko and stored them in MongoDB Atlas.
- Implemented backend API server using FastAPI and Strawberry (Python GraphQL Library).

Recent Projects

Charak — Medical Research Assistant

charak.khururu.com

Answers natural-language clinical questions with cited responses, retrieving live from PubMed and Clinical-Trials.gov v2 and charting results. [Code](#)

Bintech International

bintechint.com

Built and deployed the company website with an AI Q&A add-on; configured the custom domain, email, and DNS

Technical Skills

Languages	Python, TypeScript, JavaScript
Frontend	React, Next.js, Vite, Electron
Backend APIs	FastAPI, Express.js, GraphQL, Celery, SSE, WebSockets, Authentication/Authorization
AI & ML	PyTorch, vLLM, Ollama, IPEX-LLM, Llama-cpp, OpenVINO/OVMS, HuggingFace
Infrastructure	Docker/Compose, Nginx, Linux, GitHub Actions, OpenShell, AWS, Google Cloud
Databases	Redis, PostgreSQL, MongoDB, Qdrant
Tools	Git, Github, Gitlab, Uptime-kuma, Locust, SuperTokens, PyInstaller, Inno Setup

Education

Korea Advanced Institute of Science and Technology (KAIST)

Sep 2019 – Aug 2023

B.S. in Electrical Engineering & Computer Science

British Model College (BMC)

Jul 2015 – Jun 2017

GCE A Levels, 4A*s in Chemistry, Biology, Physics, Mathematics

Awards

Intel AI PC Innovation Challenge - First Place	NeoALI, January 2025
AI Grand Challenge, IITP - Fifth Place	NeoALI, 2023
KAIPlus+ Scholarship Award	KAIST, Apr 2023
Social Innovation Award	Tsinghua University, Jul 2021
Inclusive Leadership	Common Purpose, 2021
KAIST International Student Scholarship	KAIST, 2019-2023
Best student of the batch 2015/2017	The British College, 2017
Country Topper in AS Level Chemistry	University of Cambridge, Nov 2016